

SANAE IV BASE, Antarctica

WMO: 890040

Lat: 71.674S

Lon: 2.839W

Elev: 2680

StdP: 13.33

Time Zone: 2.00 (E02)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-28.4	-25.6	-41.2	0.6	-23.0	-38.1	0.7	-21.7	76.3	-2.7	70.3	-11.4	24.0	120	1.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.1	28.0	22.7	26.3	21.7	24.5	20.8	23.8	26.7	22.4	25.4	21.3	23.9	9.2	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.0	17.4	22.9	19.5	16.2	21.8	17.9	15.0	21.0	8.9	27.0	8.3	25.5	7.9	24.1	32.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 67.5	58.8	51.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	18.9	12.3	4.9	-1.4	-3.4	-7.1	-10.2	-10.7	-9.2	-1.5	10.0	18.7
	DBStd	12.37	3.66	5.08	5.73	7.27	8.09	7.68	7.38	7.88	7.56	7.19	5.17	4.25
	HDD50	17626	965	1055	1399	1542	1656	1714	1866	1882	1776	1598	1201	972
	HDD65	23101	1430	1475	1864	1992	2121	2164	2331	2347	2226	2063	1651	1437
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	22.8	17.2	19.8	22.9	22.0	26.9	26.4	24.2	26.4	23.6	23.6	21.9	18.0
Precipitation	PrecAvg	13.9	1.1	1.0	1.1	1.1	1.3	0.9	0.9	1.1	1.0	1.3	1.7	1.3
	PrecMax	18.6	2.6	2.2	2.9	2.0	3.0	2.8	1.9	2.4	2.7	2.7	3.6	2.9
	PrecMin	9.7	0.3	0.2	0.4	0.3	0.4	0.2	0.1	0.3	0.2	0.3	0.3	0.3
	PrecStd	2.1	0.6	0.5	0.6	0.5	0.6	0.6	0.4	0.6	0.5	0.6	0.9	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.3	27.3	19.9	18.0	17.8	13.9	10.0	13.6	12.4	15.3	25.6	32.3
		MCWB	23.3	22.2	17.9	14.9	16.7	12.4	8.7	12.4	10.2	12.9	19.1	25.3
	2%	DB	27.9	23.2	17.1	14.7	13.9	9.9	6.8	7.9	7.4	13.0	21.4	28.5
		MCWB	22.5	19.8	14.8	13.2	12.6	8.8	5.5	6.5	5.0	10.9	17.1	23.3
	5%	DB	26.2	21.4	15.1	12.5	11.2	6.9	3.8	4.5	3.8	10.8	19.1	26.7
		MCWB	21.6	18.6	13.4	11.1	10.1	5.7	2.6	3.4	2.0	9.2	16.0	22.3
	10%	DB	24.7	19.5	13.2	9.9	8.6	3.9	0.6	1.0	0.9	8.7	17.2	24.9
		MCWB	20.9	17.3	11.8	8.6	7.8	2.6	-0.4	-0.2	-0.8	7.2	14.7	21.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.0	22.5	19.0	15.8	17.0	11.8	9.1	12.0	11.2	13.4	19.7	26.8
		MCDB	28.0	25.2	19.9	17.2	17.5	12.2	9.7	12.8	11.9	14.3	23.9	30.9
	2%	WB	23.6	21.0	15.5	13.5	12.9	8.1	5.9	5.8	5.1	11.0	18.1	24.3
		MCDB	26.5	23.0	16.5	14.7	13.6	9.1	6.8	6.9	7.0	12.1	21.0	27.1
	5%	WB	22.4	18.9	13.8	11.5	9.7	5.2	2.8	2.6	1.8	9.2	16.6	22.9
		MCDB	25.2	20.5	14.8	12.8	10.4	6.3	3.8	3.6	3.4	10.4	19.0	26.1
	10%	WB	21.3	17.5	11.8	8.8	7.2	2.1	-0.2	-0.6	-0.9	6.9	15.3	21.6
		MCDB	24.0	19.3	13.2	10.0	8.0	3.3	0.8	0.5	0.5	8.2	17.3	24.4
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.2	7.8	8.3	8.2	9.1	9.2	9.0	9.1	8.6	8.1	7.7
		MCDBR	10.6	10.3	8.8	9.8	9.0	10.6	10.7	11.0	11.3	10.0	10.1	10.0
		MCWBR	8.2	8.7	7.8	9.1	8.8	9.8	10.1	11.0	9.8	8.9	8.2	7.5
	5% WB	MCDBR	9.4	8.7	7.8	9.9	9.1	10.0	10.6	11.7	10.9	9.1	9.4	9.4
		MCWBR	7.5	7.9	7.3	9.5	8.8	9.5	10.1	11.2	9.7	8.2	8.0	7.4
Clear-Sky Solar Irradiance	taub	0.220	0.206	0.180	0.122	N/A	N/A	N/A	N/A	0.120	0.164	0.185	0.204	0.221
	taud	2.225	2.241	2.233	1.981	N/A	N/A	N/A	N/A	1.950	2.178	2.203	2.219	2.203
	Ebn at Noon	326	311	282	201	0	0	0	0	196	285	318	331	331
	Edh at Noon	35	29	20	9	0	0	0	0	9	20	30	35	38
All-Sky Solar Radiation	RadAvg	2744	1791	836	187	0	0	0	0	72	554	1416	2453	3064
	RadStd	62	56	29	10	0	0	0	0	4	26	45	81	49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page